

CNAF M-3710.7

NATOPS GENERAL FLIGHT AND OPERATING INSTRUCTIONS MANUAL



CNAF M-3710.7

**DEPARTMENT OF THE NAVY
COMMANDER, NAVAL AIR FORCES**

05 MAY 2016



DEPARTMENT OF THE NAVY
COMMANDER NAVAL AIR FORCE PACIFIC
BOX 357051
SAN DIEGO, CALIFORNIA 92135-7051

COMNAVAIRFOR M-3710.7
N45
2 May 16

COMNAVAIRFOR MANUAL 3710.7

From: Commander, Naval Air Forces

Subj: NAVAL AIR TRAINING AND OPERATING PROCEDURES
STANDARDIZATION (NATOPS) GENERAL FLIGHT AND OPERATING
INSTRUCTIONS

1. Purpose. To issue policy and procedural guidance applicable to a broad spectrum of users and complement individual NATOPS manuals. This is a complete revision and should be reviewed in its entirety.
2. Background. The NATOPS program is a positive approach toward improving combat readiness and achieving a substantial reduction in the aircraft mishap rate. Standardization, based on professional knowledge and experience, provides the basis for development of sound operating procedures. The standardization program is not intended to stifle individual initiative, but rather to aid commanding officers in increasing their unit's combat potential without reducing command prestige or responsibility. With the objective of maintaining agile standardization guidance, Commander, Naval Air Forces accepted responsibility for NATOPS program management previously contained in OPNAVINST 3710.7.
3. Reproduction. Duplication of this publication for other than military use, without specific authority of Commander, Naval Air Forces, is not authorized.
4. Records Management. Records created as a result of this instruction, regardless of media or format, shall be managed per the Secretary of the Navy (SECNAV) Manual 5210.1.
5. Reports and Forms. Reports and forms required by this instruction are listed in appendix L.


P. M. SHOEMAKER

Reproduction for nonmilitary use of the information or illustrations contained in this publication is not permitted without specific approval of the issuing service (CNO).

LIST OF EFFECTIVE PAGES

TOTAL NUMBER OF PAGES IN THIS PUBLICATION IS 434, CONSISTING OF THE FOLLOWING:

<i>Page No.</i>	<i>Change</i>
1 (Reverse Blank)	0
3 (Reverse Blank)	0
5 (Reverse Blank)	0
7 (Reverse Blank)	0
9 thru 45 (Reverse Blank)	0
1-1 thru 1-5 (Reverse Blank)	0
2-1 thru 2-34	0
3-1 thru 3-24	0
4-1 thru 4-13 (Reverse Blank)	0
5-1 thru 5-36	0
6-1 thru 6-5 (Reverse Blank)	0
7-1 thru 7-7 (Reverse Blank)	0
8-1 thru 8-42	0
9-1 thru 9-3 (Reverse Blank)	0
10-1 thru 10-22	0
11-1 thru 11-17 (Reverse Blank)	0
12-1 thru 12-15 (Reverse Blank)	0
13-1 thru 13-6	0
14-1 thru 14-29 (Reverse Blank)	0
A-1 thru A-4	0
B-1 thru B-2	0
C-1 thru C-6	0
D-1 thru D-23 (Reverse Blank)	0
E-1 thru E-8	0
F-1 thru F-5 (Reverse Blank)	0
G-1 thru G-3 (Reverse Blank)	0
H-1 thru H-6	0
I-1 (Reverse Blank)	0
J-1 thru J-2	0
K-1 thru K-7 (Reverse Blank)	0
L-1 thru L-3 (Reverse Blank)	0
M-1 thru M-4	0
N-1 thru N-7 (Reverse Blank)	0
Index-1 thru Index-33 (Reverse Blank) . . .	0

INTERIM CHANGE SUMMARY

The following Interim Changes have been cancelled or previously incorporated into this manual.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE
1 thru 38	Previously incorporated or cancelled.

The following Interim Changes have been incorporated into this Change/Revision.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE
39	Multi-Subject Revisions, Including Approved Simulator Additions to Appendix K
40	UAS Policies and Operations
41	Appdx K Simulators, Life Preservers, and Waivers to Min Flt Hrs
42	Aeromedical and Swim/Physical Requirements
43	Closed Tower Ops, Weather Briefing, Parachute Warning, and Misc H-60 Simulators
44	Multi-Subject
45	P-8A Training Classification
46	Appendix K, AFAO, and Pub Reference Updates
47	Flight Approval Authority and Misc. Items

Interim Changes Outstanding — To be maintained by the custodian of this manual.

INTERIM CHANGE NUMBER	ORIGINATOR/DATE (or DATE/TIME GROUP)	PAGES AFFECTED	REMARKS/PURPOSE

RECORD OF CHANGES

Record entry and page count verification for each printed change and erratum:

[illegible]

NATOPS General Flight and Operating Instructions

CONTENTS

Page
No.

CHAPTER 1 —INTRODUCTION

1.1	GENERAL	1-1
1.1.1	Purpose and Scope	1-1
1.1.2	Submission of NATOPS Change Recommendations	1-1
1.1.3	Change Symbols	1-2
1.1.4	Areas of Responsibility	1-2
1.1.5	How To Obtain Copies	1-2
1.2	APPLICATION OF OPERATIONAL RISK MANAGEMENT ANALYSIS	1-2
1.3	OTHER GOVERNING SOURCES OF INFORMATION	1-3
1.3.1	NATOPS Manuals	1-3
1.3.2	Local Flying Rules and Instructions	1-3
1.3.3	Federal Aviation Regulations (FAR)	1-3
1.3.4	DOD Flight Information Publications (FLIPs) (NOTAL) and Notices to Airmen (NOTAMs) (NOTAL)	1-4
1.3.5	FAA Order JO 7110.65 (Air Traffic Control) (NOTAL)	1-4
1.3.6	NATOPS Air Traffic Control Manual (NAVAIR 00-80T-114)	1-4
1.3.7	DoD Detail Specification for NATOPS Program Technical Publications and Products; Style, Format, and Common Technical Content, MIL-DTL-85025B(AS)	1-4
1.3.8	Other Instructions	1-4
1.4	EXPLANATION OF TERMS	1-5
1.5	WARNINGS, CAUTIONS, AND NOTES	1-5
1.6	WORDING	1-5

CHAPTER 2 — NAVAL AIR TRAINING AND OPERATING PROCEDURES STANDARDIZATION PROGRAM

2.1	PURPOSE	2-1
2.2	NATOPS PROGRAM ORGANIZATION	2-1
2.2.1	NATOPS Program Assignments	2-1
2.2.2	Responsibilities	2-3
2.2.3	General Administrative Requirements	2-6

2.3	NATOPS PRODUCTS AND PUBLICATIONS	2-7
2.3.1	Administrative NATOPS Products and Tools	2-7
2.3.2	NATOPS Manuals and Associated Products	2-8
2.3.3	Categories of NATOPS Products	2-9
2.3.4	Changes to NATOPS Products	2-10
2.4	CREATING, UPDATING AND CANCELLING NATOPS PUBLICATIONS	2-11
2.4.1	Creating a New NATOPS	2-11
2.4.2	Updating an Existing NATOPS	2-12
2.4.3	Cancelling a NATOPS	2-13
2.5	CHANGE RECOMMENDATIONS	2-13
2.5.1	Types of NATOPS Change Recommendations	2-13
2.5.2	Submission of NATOPS Change Recommendations	2-13
2.5.3	Routine Change Recommendations	2-13
2.5.4	Interim Change Recommendations	2-14
2.5.5	Preparation and Distribution of Interim Changes	2-18
2.6	NATOPS REVIEW PROCEDURES	2-18
2.6.1	General	2-18
2.6.2	Responsibility	2-18
2.6.3	Contractor Support of NATOPS Reviews	2-20
2.6.4	Convening Decision	2-20
2.6.5	Scheduling	2-20
2.6.6	Review Location	2-20
2.6.7	Convening Announcement	2-20
2.6.8	Review Agenda	2-22
2.6.9	Pre-Reviews	2-22
2.6.10	Conduct of NATOPS Reviews	2-22
2.6.11	Review Report	2-23
2.6.12	Publication Production Package	2-29
2.6.13	Implementation of Approved Agenda Items	2-29
2.6.14	Prepublication Reviews	2-29
2.7	NATOPS EVALUATION PROCEDURES	2-30
2.7.1	General	2-30
2.7.2	Individual NATOPS Evaluations	2-30
2.7.3	Unit NATOPS Evaluation	2-34

CHAPTER 3 — POLICY GUIDANCE

3.1	POLICY CONCERNING USE OF AIRCRAFT	3-1
3.1.1	Special Policies	3-1
3.1.2	Nonessential Flights	3-2

3.1.3	Personnel Authorized To Pilot Naval Aircraft	3-3
3.1.4	Personnel Authorized To Taxi Naval Aircraft	3-4
3.1.5	Personnel Authorized To Perform Crew Duties in Naval Aircraft	3-4
3.1.6	Personnel Authorized as Project Specialists	3-5
3.2	POLICY CONCERNING USE OF SIMULATORS	3-5
3.3	ORIENTATION FLIGHTS	3-6
3.3.1	Purpose	3-6
3.3.2	Approval Authority	3-6
3.3.3	Categories of Eligible Participants for Orientation Flight	3-7
3.3.4	Orientation Flight Prerequisites	3-8
3.3.5	Flight Limitations	3-9
3.4	EMBARKATION OF PASSENGERS	3-10
3.5	FLIGHT DEMONSTRATIONS AND STATIC EXHIBITS	3-11
3.5.1	Naval Aircraft Participation	3-11
3.5.2	Approval Authority	3-11
3.5.3	Regulations	3-11
3.5.4	Exception	3-12
3.5.5	NATO Flight Demonstrations	3-12
3.5.6	NATO Live Weapons Demonstrations	3-12
3.6	EMPLOYMENT OF NAVAL AVIATORS BY CIVILIAN CONTRACTORS	3-13
3.7	COMMAND	3-13
3.7.1	Pilot in Command	3-13
3.7.2	Formation Leader	3-14
3.7.3	Mission Commander	3-14
3.7.4	Instructors	3-14
3.8	CREW RESOURCE MANAGEMENT	3-14
3.8.1	Critical Behavioral Skills	3-15
3.8.2	Effective CRM Training	3-16
3.9	OPERATIONAL-RISK MANAGEMENT	3-16
3.9.1	ORM Process Description	3-16
3.9.2	Enhancing ORM	3-17
3.10	FUNCTIONAL CHECKFLIGHTS	3-17
3.10.1	Crew Composition	3-17
3.10.2	Weather Criteria	3-17
3.11	REPORTING AND RECORDING OF DEVIATIONS AND VIOLATIONS OF FLYING REGULATIONS AND MISHAP INFORMATION	3-17
3.11.1	Reports of Investigations of Violations of Flying Regulations	3-17

		<i>Page No.</i>
3.11.2	FAA Reports and Cooperation	3-19
3.11.3	Applicability of Flying Regulations Other Than Naval	3-19
3.11.4	Alleged Offshore Air Defense Identification Zone Violations	3-19
3.11.5	Flight Personnel Training/Qualification Jacket Entry/Aviators Flight Log Book Entry ..	3-19
3.11.6	Incident Reports	3-20
3.12	CROSS-COUNTRY PLANNING	3-20
3.12.1	Cross-Country Flight	3-20
3.12.2	Risk Assessment	3-21
3.12.3	Implementation	3-21
3.13	TERMINAL INSTRUMENT PROCEDURES	3-21
3.13.1	General	3-21
3.13.2	U.S. Civil Airports	3-22
3.13.3	Other Than U.S. Airports	3-22
3.13.4	Conformance to TERPs	3-22
3.13.5	Annual Revalidation	3-22
3.14	MILITARY FLIGHT OPERATIONS QUALITY ASSURANCE (MFOQA)	3-22
3.14.1	Use of MFOQA Data	3-22
3.15	USN AVIATION SAFETY AWARENESS PROGRAM (ASAP)	3-23
3.15.1	ASAP Data Requirements	3-23
3.16	PUBLIC AIRCRAFT OPERATIONS (PAO) VERSUS CIVIL AIRCRAFT OPERATIONS	3-24
3.16.1	Airworthiness Authority	3-24
3.16.2	State Aircraft	3-24

CHAPTER 4 — FLIGHT AUTHORIZATION, PLANNING, AND APPROVAL

4.1	FLIGHT AUTHORIZATION	4-1
4.1.1	Authority	4-1
4.1.2	Documentation	4-1
4.1.3	Flightcrew Requirements	4-1
4.2	MINIMUM FLIGHTCREW REQUIREMENTS	4-1
4.2.1	Aircraft Commander Requirement	4-2
4.2.2	Insufficient NATOPS Guidance	4-2
4.2.3	Helicopters Not Requiring a Copilot	4-2
4.2.4	Use of Lookouts	4-2
4.2.5	Rescue Helicopters Operating Over Water	4-2
4.3	FLIGHT PLANNING	4-2
4.3.1	Preflight Planning	4-2

4.4	AUTHORIZED AIRFIELDS	4-3
4.4.1	Authorized Airfields for Stop-and-Go, Refueling and RON	4-3
4.4.2	Authorized Airfields and Landing Areas For Training	4-3
4.4.3	Helicopter, Tiltrotor, and VSTOL/STOL Landing Areas	4-4
4.4.4	Airfield Operations Outside Published Hours/Closed Control Tower Airfield Operations	4-4
4.4.5	Closed Airfields	4-4
4.5	FLIGHT PLANS	4-4
4.5.1	General	4-4
4.5.2	Forwarding Flight Plans to ARTCC/Flight Service Station (FSS)	4-4
4.5.3	No Communication Link	4-5
4.5.4	Direct User Access Terminal Service (DUAT)	4-5
4.5.5	Flight Plan Forms	4-5
4.5.6	Shore-to-Ship and Ship-to-Shore Operations	4-6
4.5.7	Stopover Flights Within the Contiguous United States	4-6
4.6	SUBMISSION OF THE FLIGHT PLAN	4-7
4.6.1	Pilot in Command/Formation Leader	4-7
4.6.2	Daily Flight Schedule	4-7
4.6.3	Flight Plan Approval	4-7
4.7	FLIGHT PLAN MODIFICATION	4-8
4.8	OTHER PREFLIGHT REQUIREMENTS	4-8
4.8.1	Call Sign Requirements	4-8
4.8.2	Manifest Requirements	4-8
4.8.3	Flight Route Weather Briefing	4-8
4.8.4	Weather Criteria for Filing	4-10
4.8.5	Minimum Fuel Requirements	4-11
4.8.6	Weight and Balance Control	4-12
4.9	CLOSING OF FLIGHT PLAN	4-13
4.9.1	Military Installations	4-13
4.9.2	Nonmilitary Installations	4-13

CHAPTER 5 — FLIGHT RULES

5.1	GENERAL FLIGHT RULES	5-1
5.1.1	Aircraft Lighting	5-1
5.1.2	Right-of-Way Between Single and Formations of Aircraft	5-2
5.1.3	Unusual Maneuvers Within Class B, C, or D Airspace	5-2
5.1.4	Aircraft Speed	5-2

5.1.5	Special Use Airspace	5-3
5.1.6	Military Training Routes (MTRs)	5-4
5.1.7	Flight Over the High Seas	5-6
5.1.8	Supersonic Flight Operations	5-7
5.1.9	Aerobatic Flight	5-8
5.1.10	Simulated Air Combat Maneuvering (ACM) Training Rules	5-8
5.1.11	Simulated Instrument Flight	5-18
5.1.12	Formation Flying	5-19
5.2	VISUAL FLIGHT RULES PROCEDURES	5-21
5.2.1	Compliance With Directives	5-21
5.2.2	Judgment	5-21
5.2.3	See and Avoid	5-21
5.2.4	Weather Minimums	5-21
5.2.5	Weather Conditions Precluding VFR Flight	5-22
5.2.6	Additional Requirements	5-22
5.3	INSTRUMENT FLIGHT RULES AND POSITIVE CONTROL PROCEDURES	5-23
5.3.1	General Requirements	5-23
5.3.2	Aircraft Equipment Requirements	5-25
5.3.3	RNAV/GPS Navigation	5-28
5.3.4	Instrument Departures	5-28
5.3.5	Instrument Approaches and Landing Minimums	5-29
5.4	HELICOPTER/TILTROTOR OPERATIONS	5-31
5.4.1	Helicopter/Tiltrotor Operations in Class B, C, or D Airspace	5-31
5.4.2	Helicopter/Tiltrotor Terrain Flight Operations	5-32
5.4.3	Helicopter/Tiltrotor Night Hover Operation Over Water	5-32
5.4.4	Helicopter Operations	5-32
5.5	REDUCING FLIGHT-RELATED DISTURBANCES	5-32
5.5.1	Annoyance to Civilians and Endangering Private Property	5-32
5.5.2	Disturbance of Wildlife	5-33
5.5.3	Zooming of Vessels	5-33
5.5.4	Avoidance of Commercial Carriers and Aircraft of Civil Registry	5-33
5.5.5	Avoidance of Installations Important to Defense	5-34
5.5.6	Jettisoning Fuel	5-34
5.5.7	Air-to-Air Missile Training Flights	5-34
5.5.8	Expenditure of Airborne Stores Through Extensive Cloud Cover	5-34
5.6	FLAMEOUT APPROACHES	5-35
5.6.1	Actual Flameout Approaches	5-35
5.6.2	Simulated Flameout Approaches	5-35

5.7	FLIGHT OPERATIONS WITH NIGHT VISION DEVICES	5-35
5.7.1	General	5-35
5.7.2	Operating Limitations	5-35

CHAPTER 6 — AIR TRAFFIC CONTROL

6.1	APPLICABILITY	6-1
6.2	AIR TRAFFIC CONTROL PROCEDURES	6-1
6.2.1	Authorized Personnel	6-1
6.2.2	Control Tower	6-1
6.2.3	Control of Formation Flights	6-1
6.2.4	Taxi Instructions	6-1
6.2.5	Departure Instructions	6-2
6.2.6	Minimum Fuel	6-2
6.2.7	Handling of VIP Aircraft	6-2
6.2.8	Approach Instructions	6-3
6.3	LANDING INSTRUCTIONS	6-3
6.3.1	Reduced Same Runway Separation	6-3
6.3.2	Procedure for Checking Wheels Down and Locked	6-4
6.3.3	Runway Braking Action Advisory/Condition Readings	6-4
6.4	LETTERS OF AGREEMENT	6-4
6.5	VITAL MILITARY OPERATIONS	6-4
6.5.1	Priority	6-4
6.5.2	Letters of Agreement	6-5
6.6	AIRFIELD VEHICLE OPERATIONS	6-5
6.6.1	Airfield Vehicle Operators Course (AVOC)	6-5

CHAPTER 7 — SAFETY

7.1	FLIGHT PRECAUTION	7-1
7.1.1	General Precautions	7-1
7.1.2	Starting, Turning, and Taxiing	7-2
7.1.3	Takeoff	7-4
7.1.4	Takeoff and Landing Checklists	7-4
7.1.5	Engine Malfunctions on Multiengine Aircraft	7-5
7.1.6	Distress and Emergency	7-5
7.1.7	Ditching and Bailout	7-5
7.1.8	Command and Control Communication	7-6
7.1.9	Tobacco Products in Aircraft	7-6
7.2	SAFETY BELTS AND SHOULDER HARNESSSES	7-6
7.3	UNUSUAL PERFORMANCE OF AIRCRAFT	7-7

CHAPTER 8 — AEROMEDICAL AND SURVIVAL

8.1	GENERAL	8-1
8.2	AVIATION LIFE SUPPORT SYSTEMS	8-1
8.2.1	Aircrew Personal Protective Equipment Requirements	8-1
8.2.2	Liferafts	8-8
8.2.3	Parachutes	8-8
8.2.4	Oxygen/Cabin Pressurization	8-8
8.2.5	Chemical, Biological, Radiological, Nuclear, Defense (CBRND) Protective Equipment	8-13
8.3	HUMAN PERFORMANCE AND AEROMEDICAL QUALIFICATIONS FOR FLIGHT AND FLIGHT SUPPORT	8-13
8.3.1	General	8-13
8.3.2	Factors Affecting Aircrew Performance	8-13
8.3.3	Performance Maintenance During Continuous and Sustained Operations	8-20
8.4	NAVAL AVIATION SURVIVAL TRAINING PROGRAM	8-21
8.4.1	Training Requirements	8-23
8.4.2	Records	8-26
8.4.3	Physical and Training Prerequisites for Participation in NASTP Training	8-26
8.4.4	Training Waivers/Qualification Extensions	8-29
8.4.5	Coordination	8-30
8.4.6	Approved Curricula	8-30
8.4.7	Grading	8-34
8.4.8	Environmental Exposure	8-35
8.5	FLEET AIR INTRODUCTION/LIAISON OF SURVIVAL AIRCREW FLIGHT EQUIPMENT (FAILSAFE) PROGRAM	8-35
8.6	NVD TRAINING PROGRAM	8-35
8.7	SEARCH AND RESCUE PILOT/RESCUE SWIMMER TRAINING	8-36
8.7.1	Definitions	8-36
8.7.2	Training Requirements	8-37
8.7.3	Prerequisites	8-37
8.8	AVIATION PHYSICAL EXAMINATIONS AND QUALIFICATIONS	8-37
8.8.1	General Requirements	8-37
8.8.2	Required Evaluations	8-37
8.8.3	Scope of Examinations	8-39
8.8.4	Disposition of Aircrew Found Not Physically Qualified (NPQ)	8-39
8.8.5	Medical Service Groups	8-41
8.8.6	Medical Requirements for Contract Civilians Operating Naval Aircraft	8-42

CHAPTER 9 — MISCELLANEOUS

9.1	PARACHUTE JUMPS	9-1
9.1.1	General	9-1
9.1.2	Delayed Release Jumps	9-1
9.1.3	Jump Precautions	9-1
9.1.4	Federal Aviation Regulations	9-1
9.1.5	Demonstrations	9-1
9.2	SECURITY OF AIRCRAFT AWAY FROM BASE	9-1
9.2.1	General	9-1
9.2.2	Aircraft Mishap	9-1
9.3	AIRCRAFT FUEL PURCHASE	9-2
9.4	AIRCRAFT NOISE ABATEMENT	9-2
9.5	CLAIMS FOR PERSONAL PROPERTY IN MARITIME DISASTERS OF AIRCRAFT	9-2
9.6	U.S. CUSTOMS, HEALTH, IMMIGRATION, AND AGRICULTURAL CLEARANCE	9-2
9.6.1	Naval Aircraft	9-2
9.6.2	Military Aircraft Arriving in the Continental U.S. From Overseas	9-2
9.6.3	Discharging of Passengers/Cargo	9-2
9.6.4	Foreign Military Aircraft	9-3
9.6.5	Medical or Economic Insect Pests	9-3
9.7	DISPERSAL OF PESTICIDES	9-3
9.8	AIRCRAFT TAIL LETTERS AND SIDE NUMBERS	9-3

CHAPTER 10 — FLIGHT RECORDS, REPORTS, AND FORMS

10.1	NAVAL FLIGHT RECORD SUBSYSTEM	10-1
10.2	AIRCRAFT INSPECTION AND ACCEPTANCE (AIA) RECORD, OPNAV 4790/141	10-1
10.2.1	Pilot in Command	10-1
10.2.2	“Limitations/Remarks” Section	10-1
10.3	NAVAL AIRCRAFT FLIGHT RECORD, OPNAV 3710/4	10-1
10.3.1	Documentation of the Naval Aircraft Flight Record	10-2
10.3.2	Aircraft Data Section	10-6
10.3.3	Aircrew Data Section	10-9

		<i>Page No.</i>
10.3.4	Logistics Data Section	10-12
10.3.5	Weapons Proficiency Data Section	10-13
10.3.6	Personnel Data	10-15
10.3.7	Personnel Exchange Program/DCMA/Any Aeronautically Designated Personnel Assigned to an Activity Where NDCSC Support Is Not Available	10-16
10.3.8	Civilian Crewmembers Flying Naval Aircraft (Active)	10-17
10.3.9	Fleet Readiness Centers	10-17
10.4	RECORD OF COMPLETED FLIGHT TIME (OPNAV 3760/37)	10-17
10.5	MASTER FLIGHT FILES	10-17
10.5.1	Specific Requirements	10-17
10.5.2	Procedures for Maintaining Master Flight Files	10-17
10.5.3	Master Flight File Certification	10-18
10.5.4	Retention of Master Flight Files	10-19
10.6	AVIATORS FLIGHT LOG BOOK, OPNAV 3760/31	10-19
10.6.1	General Policies	10-19
10.6.2	Entries	10-19
10.7	NATOPS FLIGHT PERSONNEL TRAINING/QUALIFICATION JACKET, OPNAV 3760/32	10-22
10.8	MONTHLY INDIVIDUAL FLIGHT ACTIVITY REPORT (NAVFLIRS-3)	10-22
10.9	INDIVIDUAL FLIGHT ACTIVITY REPORTING SYSTEM (IFARS)	10-22
10.9.1	Background	10-22

CHAPTER 11 — GENERAL INSTRUCTIONS ON DUTY INVOLVING FLYING AND ANNUAL FLIGHT PERFORMANCE REQUIREMENTS

11.1	SCOPE, PURPOSE, AND APPLICABILITY	11-1
11.1.1	General Policies	11-1
11.2	OPERATIONAL FLYING	11-2
11.2.1	Aeromedical Officer Flying Policy	11-2
11.2.2	Aviation Qualified Foreign Area Officer (AFAO) Policy	11-3
11.2.3	Aviation Operations Officer (AVOPS)	11-3
11.2.4	Additional Ratings	11-3
11.2.5	Minimum Flying Hours	11-4
11.2.6	Prorating Minimums	11-5
11.2.7	Aviation Qualification/Currency Requirements Summary	11-5
11.2.8	Flying Activity Denied	11-5
11.2.9	Policy Governing Assignment of Inactive Reserve Personnel	11-10
11.3	AVIATION CAREER INCENTIVE PAY	11-10
11.3.1	Definitions	11-10

11.3.2	Policy and Procedures	11-10
11.3.3	Aviation Career Incentive Pay for Rated Members (Rated Members Include Aeronautically Designated Naval Aviators and Naval Flight Officers)	11-11
11.4	ENLISTED CREWMEMBERS	11-11
11.4.1	Navy Crewmembers	11-11
11.4.2	Marine Corps Crewmembers	11-13
11.4.3	Hazardous Duty Incentive Pay for Enlisted Member/Aeronautically Designated Enlisted and Nondesignated Officers	11-14
11.5	WAIVERS OF MINIMUM FLYING REQUIREMENTS	11-14
11.5.1	Authority to Waive	11-14
11.5.2	Action Required	11-14
11.5.3	Assignment of Other Than Permanently Designated Aeronautical Personnel	11-16
11.6	POLICY GOVERNING LOGGING, REPORTING, AND USE OF SIMULATOR TIME	11-16
11.6.1	Policy Governing Flying Time Substitution	11-16
11.6.2	Policy Governing NATOPS Evaluation Flight Substitution	11-17
11.7	INDIVIDUAL AND COMMAND RESPONSIBILITIES	11-17
11.7.1	Supervision	11-17
11.7.2	Responsibilities	11-17
11.8	REVOCATION OF ORDERS TO DUTY INVOLVING FLYING	11-17

CHAPTER 12 — CLASSIFICATION AND QUALIFICATION OF FLIGHT PERSONNEL

12.1	SCOPE	12-1
12.2	MULTIPILOTED FIXED-WING AIRCRAFT (PILOT)	12-1
12.2.1	Pilot Classification	12-1
12.2.2	Specific Requirements for Qualification	12-1
12.2.3	General Requirements for Qualification	12-3
12.3	MULTIPILOTED ROTARY-WING AIRCRAFT (PILOT)	12-4
12.3.1	Pilot Classification	12-4
12.3.2	Specific Requirements for Qualification	12-4
12.3.3	General Requirements for Qualification	12-6
12.4	MULTIPILOTED TILTROTOR AIRCRAFT (PILOT)	12-6
12.4.1	Pilot Classification	12-6
12.4.2	Specific Requirements for Qualifications	12-6
12.4.3	Initial Qualification	12-8
12.4.4	Requalification	12-8
12.4.5	Time Limits	12-8

12.5	NAVAL FLIGHT OFFICERS	12-9
12.5.1	Naval Flight Officer Classification	12-9
12.5.2	Specific Requirements for Qualification	12-9
12.5.3	General Requirements for Qualification	12-10
12.6	MARINE AERIAL NAVIGATION OFFICER	12-11
12.7	QUALIFICATIONS OF UAS FLIGHTCREW	12-11
12.8	TRAINING OF ENLISTED FLIGHT PERSONNEL	12-11
12.8.1	General	12-11
12.8.2	Flight Records	12-11
12.8.3	Auditing of Enlisted Flight Record	12-11
12.8.4	Allocation of Temporary Flight Orders	12-11
12.9	CLASSIFICATION AND QUALIFICATION OF NAVAL AIRCREWMAN	12-12
12.9.1	Naval Aircrewman Classification	12-12
12.9.2	General Requirements for Positional Qualification as a Naval Aircrewman	12-12
12.9.3	Proficiency	12-12
12.9.4	Maximum Time Limit for Positional Qualification as Naval Aircrewman	12-12
12.9.5	Time of Requalification for Naval Aircrewman	12-12
12.9.6	Qualification Waivers for Naval Aircrewmembers	12-13
12.10	QUALIFYING AUTHORITIES	12-13
12.10.1	Aeronautical Organizations	12-13
12.10.2	Non-aeronautical Organizations	12-13
12.10.3	Fleet Replacement Squadrons	12-13
12.10.4	Guidance for Qualifying Authorities	12-13
12.11	QUALIFICATION TO TRANSITION INTO JET, HELICOPTER, OR TILTROTOR AIRCRAFT	12-14
12.11.1	Minimum Training Syllabus Requirements	12-14
12.11.2	Action	12-15
12.11.3	Chief of Naval Air Training Responsibility	12-15
12.12	REPORTS	12-15
12.12.1	Navy Flight Personnel	12-15
12.12.2	Marine Corps Flight Personnel	12-15
12.12.3	Revocation of Qualifications	12-15

CHAPTER 13 — INSTRUMENT RATINGS AND QUALIFICATIONS

13.1	INSTRUMENT RATINGS AND QUALIFICATIONS	13-1
13.1.1	Pilots/Naval Flight Officers Required To Maintain Instrument Ratings/Qualifications	13-1
13.1.2	Renewal/Expiration of Instrument Ratings and Qualifications	13-1
13.1.3	Composition and Functions of Instrument Flight Boards	13-3

13.2	REQUIREMENT FOR INSTRUMENT RATINGS	13-3
13.2.1	Standard Rating	13-3
13.2.2	Special Rating	13-4
13.2.3	Failure To Meet Requirements	13-4
13.3	INSTRUMENT RATING FORMS	13-5
13.4	AIRCRAFT CONSIDERATIONS	13-5
13.5	GPS NAVIGATION TRAINING	13-5
13.5.1	General	13-5
13.5.2	Ground Instruction	13-5
13.5.3	GPS Navigation Flight Training	13-6

CHAPTER 14 — UAS POLICIES AND OPERATIONS

14.1	PURPOSE AND SCOPE	14-1
14.1.1	Directives, Procedures and Terminology Applicable to UAS Operations	14-1
14.1.2	Military Reporting Procedures For UAS Flight Deviations	14-1
14.1.3	Federal Aviation Regulations (FAR)	14-1
14.1.4	Other UAS-Specific Publications	14-1
14.1.5	Certificates of Authorization (COAs)	14-2
14.1.6	Compliance with UAS-Related Directives	14-2
14.1.7	Explanation of UAS Groups	14-3
14.1.8	Waivers	14-3
14.1.9	Non-Programs of Record (POR) Systems	14-4
14.2	UAS NATOPS PROGRAM	14-4
14.2.1	UAS Qualification (BUQ) Levels	14-4
14.2.2	UAS NATOPS Program Implementation	14-4
14.3	POLICY GUIDANCE FOR USE OF UASs	14-4
14.3.1	Guidance for the Domestic Use of UAS	14-4
14.3.2	General Operating Precautions	14-5
14.3.3	Prohibited Maneuvers	14-5
14.3.4	Displays and Demonstrations	14-5
14.3.5	UAS Command Responsibilities	14-5
14.3.6	Transfer of UAS Responsibilities During Flight	14-6
14.3.7	Non-Participating Personnel	14-7
14.3.8	Functional Checkflights (FCFs)	14-7
14.4	FLIGHT AUTHORIZATION AND PLANNING	14-7
14.4.1	Requirement for Flight Authorization	14-7

		<i>Page No.</i>
14.4.2	UAS Flightcrew Requirements	14-8
14.4.3	Positive Control Requirement	14-9
14.4.4	UAS Preflight Planning	14-9
14.4.5	Airspace	14-10
14.4.6	Airfield Requirements	14-10
14.4.7	Weather Planning	14-11
14.4.8	Mission Essential Subsystem Matrix (MESM)	14-12
14.4.9	Fuel and Propulsion Battery Charge Considerations	14-13
14.4.10	Risk Assessment	14-13
14.4.11	UAS Preflight Briefings	14-13
14.5	GENERAL FLIGHT RULES	14-15
14.5.1	Right-of-way Rules	14-15
14.5.2	Airspace	14-16
14.5.3	UAS VFR Requirements	14-16
14.5.4	UAS IFR Requirements	14-17
14.5.5	UAS Airport Operating Procedures	14-18
14.6	AIR TRAFFIC CONTROL	14-18
14.7	SAFETY	14-18
14.7.1	Conduct of Flight	14-18
14.7.2	Starting, Turning, and Taxiing	14-18
14.8	AEROMEDICAL REQUIREMENTS	14-19
14.8.1	Aviation Physical Examinations and Qualifications	14-19
14.8.2	Human Performance and Aeromedical Qualifications For Flight and Flight Support Personnel	14-19
14.9	MISCELLANEOUS	14-20
14.10	FLIGHT RECORDS, REPORTS AND FORMS	14-20
14.10.1	Naval Flight Record Subsystem (NAVFLIRS)	14-20
14.10.2	Flight Records for UAS Groups 1 and 2	14-20
14.10.3	Groups 3 to 5 UAS Flight Logs	14-21
14.11	GENERAL INSTRUCTIONS ON DUTY INVOLVING FLYING AND ANNUAL FLIGHT PERFORMANCE REQUIREMENTS	14-22
14.11.1	Operating UAS While in a Leave Status	14-22
14.11.2	Operational Flying	14-22
14.11.3	Operating Proficiency	14-22
14.11.4	UAS Simulators	14-23
14.12	USAC CLASSIFICATIONS AND DESIGNATION OF UAS FLIGHT AND MAINTENANCE PERSONNEL	14-24
14.12.1	UASC Classifications	14-24
14.12.2	Basic UAS Qualification (BUQ) Levels	14-24
14.12.3	Specific Requirements for Qualification	14-24

14.12.4	General Requirements for Qualification	14-25
14.12.5	Failure To Qualify	14-26
14.12.6	Revocation of Qualifications	14-26
14.13	UAS INSTRUMENT RATING	14-26
14.13.1	Requirement For UAS Instrument Ratings	14-26
14.13.2	UAS Instrument Rating Annual Qualification Requirements	14-27
14.13.3	Instrument Rating Administration	14-27
14.13.4	Initial Qualification For UAS Instrument Ratings	14-28
14.13.5	UAS Instrument Rating Renewal Requirements	14-28
14.13.6	Failure To Meet Renewal Requirements	14-29
14.13.7	Revoking of Instrument Ratings	14-29

APPENDIX A — NATOPS FLIGHT PERSONNEL TRAINING/QUALIFICATION JACKET

A.1	INTRODUCTION	A-1
A.1.1	Purpose	A-1
A.1.2	Scope	A-1
A.1.3	Responsibility	A-1
A.1.4	Security	A-1
A.1.5	Disposition	A-1
A.1.6	Review	A-3
A.1.7	Design	A-3
A.1.8	Maintenance	A-3
A.1.9	Forms	A-3
A.2	ASSEMBLY AND MAINTENANCE	A-3
A.2.1	General	A-3
A.2.2	Qualifications and Achievements	A-4
A.2.3	Training	A-4
A.2.4	Flight Records	A-4
A.2.5	Procurement	A-4

APPENDIX B — AIRCRAFT VISUAL IDENTIFICATION SYSTEM

B.1	GENERAL	B-1
B.1.1	Unit Identification	B-1
B.1.2	Aircraft Side Numbers	B-2
B.1.3	Marking of Aircraft	B-2

APPENDIX C — DIRECTIVES LISTED AS REFERENCES IN THIS MANUAL

C.1	REFERENCES TO DIRECTIVES	C-1
C.1.1	Directives Referenced in this Manual	C-1
C.1.2	Websites for the Above Directives	C-5

APPENDIX D — TOTAL MISSION REQUIREMENT (TMR) CODES

D.1	PURPOSE	D-1
D.2	NAVAL AIRCRAFT/SIMULATOR FLIGHT CLASSIFICATION SYSTEM	D-1
D.2.1	Primary Source	D-1
D.2.2	Deviation	D-1
D.3	APPLICABILITY OF THE TOTAL MISSION REQUIREMENT CODES	D-1
D.4	CLASSIFICATION OF TOTAL MISSION REQUIREMENT CODES	D-1
D.4.1	Purpose of Flight	D-1
D.5	GENERAL/SPECIFIC PURPOSE OF FLIGHT CODE COMBINATIONS A THROUGH I (TRAINING FLIGHTS)	D-2
D.5.1	General Purpose Codes	D-2
D.5.2	Specific Purpose Codes	D-3
D.6	GENERAL/SPECIFIC PURPOSE OF FLIGHT CODE COMBINATIONS J THROUGH R (SERVICE FLIGHTS)	D-3
D.6.1	SPCs To Be Used With GPCs J and K for Service Flights	D-3
D.6.2	GPCs L, M, N, and O for Service Flights	D-4
D.6.3	SPCs Used With GPC P	D-5
D.6.4	SPCs Used With GPC Q	D-6
D.6.5	SPCs Used With GPC R	D-6
D.7	GENERAL/SPECIFIC PURPOSE OF FLIGHT CODE COMBINATIONS S THROUGH Z (COMBAT FLIGHTS)	D-7
D.8	CURRENTLY ASSIGNED TOTAL MISSION REQUIREMENT CODES	D-9

APPENDIX E — NAVAL AVIATION SURVIVAL TRAINING PROGRAM (NASTP) REQUIREMENTS

E.1	PURPOSE	E-1
E.2	NASTP TRAINING STATUS (DOES NOT INCLUDE NON-AIRCRAFT SPECIFIC NASTP TRAINING COURSES LISTED IN FIGURE E-1)	E-1
E.3	NASTP TRAINING REQUIREMENTS	E-2
E.4	NASTP CURRICULA OUTLINE	E-3
E.5	AIRCRAFT CLASS	E-5
E.6	APPROVED NASTP TRAINING SITES (AVIATION SURVIVAL TRAINING CENTERS (ASTC))	E-6
E.7	NASTP ADJUNCTIVE TRAINING TOPIC GUIDE	E-6
E.8	NASTP TRAINING QUALIFICATION LETTER FOR AIRCRAFT SPECIFIC TRAINING	E-8

APPENDIX F — EXCEPTION, SPECIAL QUALIFICATION, SERVICE, LANDING, AND APPROACH CODES

F.1	PURPOSE	F-1
F.2	EXCEPTION CODES	F-1
F.3	SPECIAL QUALIFICATION CODES	F-1
F.4	SERVICE CODES	F-3
F.5	LANDING CODES	F-3
F.6	APPROACH CODES	F-4

APPENDIX G — TIME ZONE, SYSTEM STATUS, PASSENGER PRIORITY, AND OPPORTUNE CARGO CODES

G.1	PURPOSE	G-1
G.2	TIME ZONE CODES	G-1
G.3	SYSTEM STATUS CODES	G-2
G.4	PASSENGER PRIORITY CODES	G-2
G.5	OPPORTUNE CARGO CODES	G-2

APPENDIX H — WEAPONS PROFICIENCY CODES

H.1	PURPOSE	H-1
H.2	ORDNANCE CODES	H-1
H.3	ORDNANCE DELIVERY DATA CODES	H-4
H.3.1	Ordnance System/Automatic Delivery Code	H-4
H.3.2	Manual Delivery Code	H-5
H.4	MISCELLANEOUS DATA RECORD CODES	H-6

APPENDIX I — SUPPORT CODES

I.1	PURPOSE	I-1
-----	---------------	-----

APPENDIX J — USMC PERSONAL DATA SYLLABUS AND STATUS CODES

J.1	PURPOSE	J-1
J.2	USMC ASSIGNED SYLLABUS (TEC) CODES	J-1
J.3	USMC SYLLABUS STATUS (SSC) CODES	J-1
J.4	USMC AIRCREW STATUS (ASC) CODES	J-2

APPENDIX K — COMNAVAIRFOR APPROVED IFAR SIMULATORS

K.1	PURPOSE	K-1
K.2	DESIGNATION OF NEW SIMULATORS	K-1
K.3	NAVY SIMULATORS (PILOT AND NFO SPECIAL CREW TIME)	K-2
K.4	NAVY SIMULATORS (SPECIAL CREW TIME ONLY)	K-5
K.5	NON-NAVY SIMULATORS (PILOT AND SPECIAL CREW TIME)	K-5

APPENDIX L — LIST OF FORMS AND REPORTS

L.1	PURPOSE	L-1
L.2	FORMS	L-1
L.3	FORMS WEBPAGES	L-3

APPENDIX M — STANDARDIZED ACM TRAINING RULES BRIEFING GUIDE

M.1	PURPOSE	M-1
M.2	ACM TRAINING RULES	M-1
M.2.1	Administrative	M-1

APPENDIX N — BUQ LEVELS

N.1	PURPOSE	N-1
N.2	BUQ LEVELS I-IV TRAINING	N-1

INDEX	Index-1
--------------------	---------

LIST OF ILLUSTRATIONS

Page
No.

CHAPTER 1 —INTRODUCTION

Figure 1-1.	CNAF M-3710.7 Areas of Responsibility	1-2
-------------	---	-----

CHAPTER 2 — NAVAL AIR TRAINING AND OPERATING PROCEDURES STANDARDIZATION PROGRAM

Figure 2-1.	NATOPS Program Organization	2-2
Figure 2-2.	NATOPS Flight Manual Waiver Authority	2-8
Figure 2-3.	Interim Change Recommendation Approval Process	2-15
Figure 2-4.	Sample NATOPS Interim Change Recommendation Message	2-16
Figure 2-5.	Sample NATOPS Interim Change Message	2-19
Figure 2-6.	Sample NATOPS Review Convening Message	2-21
Figure 2-7.	Sample NATOPS Review Report Cover Letter	2-25
Figure 2-8.	Sample NATOPS Review Agreement	2-26
Figure 2-9.	Sample Pilot/NFO/Enlisted Aircrew Flight Logbook Entry	2-33

CHAPTER 4 — FLIGHT AUTHORIZATION, PLANNING, AND APPROVAL

Figure 4-1.	IFR Filing Criteria	4-10
-------------	---------------------------	------

CHAPTER 5 — FLIGHT RULES

Figure 5-1.	ACM Qualification	5-10
Figure 5-2.	ACM Currency	5-11
Figure 5-3.	Basic VFR Flight Minimums	5-24

CHAPTER 8 — AEROMEDICAL AND SURVIVAL

Figure 8-1.	Antiexposure Suit Requirements	8-3
Figure 8-2.	Wind Chill Index	8-5
Figure 8-3.	Occurrence of Major Injury Resulting from Parachute Landing on Land or Ship in Ejections Between 1969 and 1998 in High Winds	8-9
Figure 8-4.	Unpressurized Aircraft with Oxygen Systems Available	8-10
Figure 8-5.	AFRL DCS Risk Assessment Model (No Pre-Oxygenation and Mild Activity)	8-10
Figure 8-6.	AFRL DCS Risk Assessment Model (100% Pre-Oxygenation and Mild Activity)	8-11
Figure 8-7.	Oxygen Requirement for Pressurized Aircraft Other Than TACTICAL Jet Aircraft	8-12
Figure 8-8.	Maximum Recommended Flight Time	8-15
Figure 8-9.	COMNAVAIRSYSCOM Certified Crewmember Weights for Ejection Seat Aircraft	8-21
Figure 8-10.	COMNAVAIRSYSCOM Certified Crewmember Weights for Crash Force Attenuating Seat Aircraft	8-22
Figure 8-11.	Medical Recommendation For Flying or Special Operational Duty	8-27

CHAPTER 10 — FLIGHT RECORDS, REPORTS, AND FORMS

Figure 10-1.	Aircraft Data Section (OPNAV 3710/4)	10-7
Figure 10-2.	Aircrew Data Section (OPNAV 3710/4)	10-9
Figure 10-3.	Logistics Data Section (OPNAV 3710/4)	10-12
Figure 10-4.	Weapons Proficiency Data Section (OPNAV 3710/4)	10-14
Figure 10-5.	Personnel Data Section (OPNAV 3710/4)	10-15

CHAPTER 11 — GENERAL INSTRUCTIONS ON DUTY INVOLVING FLYING AND ANNUAL FLIGHT PERFORMANCE REQUIREMENTS

Figure 11-1.	Aviation Qualification/Currency Requirements Summary (Naval Aviator)	11-6
Figure 11-2.	Aviation Qualification/Currency Requirements Summary (NFO/AVOPS/Flight Surgeon)	11-7
Figure 11-3.	Aviation Qualification/Currency Requirements Summary (Naval Aircrewman (NAC))	11-8
Figure 11-4.	Aviation Status Indicator Codes	11-12
Figure 11-5.	Minimum Flight Time Requirements Waiver Request	11-15

CHAPTER 14 — UAS POLICIES AND OPERATIONS

Figure 14-1.	UAS Groups	14-3
--------------	------------	------

APPENDIX A — NATOPS FLIGHT PERSONNEL TRAINING/QUALIFICATION JACKET

Figure A-1.	Privacy Act Statement	A-2
-------------	-----------------------	-----

APPENDIX C — DIRECTIVES LISTED AS REFERENCES IN THIS MANUAL

Figure C-1.	Directives Listed As References in this Manual	C-1
Figure C-2.	Websites Hosting Directives Listed in this Manual	C-5

APPENDIX D — TOTAL MISSION REQUIREMENT (TMR) CODES

Figure D-1.	Total Mission Requirement (TMR) Codes	D-9
-------------	---------------------------------------	-----

APPENDIX E — NAVAL AVIATION SURVIVAL TRAINING PROGRAM (NASTP) REQUIREMENTS

Figure E-1.	NASTP Training Status	E-1
Figure E-2.	NASTP Courses	E-2
Figure E-3.	New Versus Previous NASTP Curricula Nomenclature	E-3
Figure E-4.	Aircraft Class	E-5

APPENDIX L — LIST OF FORMS AND REPORTS

Figure L-1.	Forms Referenced in this Manual	L-1
Figure L-2.	Webpages Hosting Forms Referenced in this Manual	L-3

APPENDIX N — BUQ LEVELS

Figure N-1.	BUQ Level I	N-1
Figure N-2.	BUQ Level II	N-4
Figure N-3.	BUQ Level III	N-5
Figure N-4.	BUQ Level IV	N-6

GLOSSARY

The explanation or definitions of terms and abbreviations commonly used in the aviation community can be found in FAR, Part 1, and DOD FLIP General Planning, Chapter 2; and Aeronautical Information Manual (AIM) Pilot/Controller Glossary. No effort to duplicate these terms is intended. Where terms are used in this instruction with a different connotation or where definitions are lacking in the above-mentioned publications, the explanations of such terms are included.

A

Actual Instrument Approach. When actual instrument conditions are encountered below 1,000 feet above the airport/flight deck elevation during an instrument approach.

Actual Instrument Conditions. Conditions external to the aircraft in flight that do not permit visual reference to the horizon.

Aerobatic Flight Maneuvers. An intentional maneuver involving an abrupt change in aircraft attitude, intentionally performed spins, or other maneuvers requiring pitch/dive angles greater than 45°, bank angles greater than 60°, or accelerations greater than 2 gs. A maneuver that conforms to the model NATOPS manual (e.g., break, weapons delivery, autorotations, etc.) is not considered to be aerobatic flight.

Aeromedical Dual Designator. An aeronautically designated Medical Department officer (i.e., flight surgeon, aerospace physiologist, aerospace experimental psychologist, or aviation optometrist) with the Additional Qualification Designator (AQD) of either 6AC (Med Dept & NFO) or 6AE (Med Dept & Pilot).

Aeromedical Officer. An aeronautically designated Medical Department officer (i.e., flight surgeon, aerospace physiologist, aerospace experimental psychologist, or aerospace optometrist), or officer student in a course of instruction leading to such designation.

Aeronautically Designated Personnel. A collective term that applies to all Naval Aviators, Naval Flight Officers, Naval Aerial Observers (USMC), Naval Flight Surgeons, Naval Aerospace Physiologists, Naval Aerospace Experimental Psychologists, Aviation Operations Officers (AVOPS), Aviation Qualified Foreign Area Officers (AFAO), Aviation Warfare Systems Operator (AW rating), personnel assigned by the Chief of Naval Personnel under a distribution Naval Enlisted Classification (NEC) of 82XX and 94XX, and USMC-enlisted crewmembers. Enlisted noncrewmembers are not considered aeronautically designated.

Aircraft Class. A broad classification as to the general mission purpose of an aircraft design (e.g., attack, fighter, helicopter, patrol, transport, vertical takeoff and landing and unmanned aerial vehicles).

Aircraft Commander Time. The individual flight time during which an individual, designated as a qualified aircraft commander in the aircraft model being flown, is serving as pilot in command. Aircraft commander time is a measure of command experience rather than of pilot experience.

Aircraft Model. The basic mission symbol and design number (e.g., P-3, S-3, F/A-18, and H-60).

Aircraft Series. The specific version of aircraft within the same model (e.g., AV-8B; H-46D or E; F/A-18D or E/F).

Aircraft Type. The broadest classification of aircraft as to physical characteristics (i.e., fixed-wing, rotary-wing or tilt-rotor).

Aircrew. A collective term that applies to all categories of personnel in a flight status either as crew or noncrewmember. Aircrew are military personnel on competent flight orders or civilian personnel whose duties require frequent and regular participation in aerial flights to perform inflight

functions such as installation, maintenance, evaluation of airborne technical equipment (maintenance skins), communication specialists, photo specialists, etc.

Aviation Qualified Foreign Area Officers (AFAO). Foreign Area Officers previously designated as naval aviators (pilots) and awarded the Aviation Qualified FAO AQD of FFQ.

Aviation Training System (ATS). The ATS is used to manage Marine Aviation Training (Maintenance, Aircrew and Command and Control) by facilitating standardization, evaluation and Crew Resource Management in order to provide a tactically relevant training continuum.

B

Bolter. An attempted arrested landing on a carrier in which some portion of the aircraft, such as the landing gear or hook, touches the deck but the arresting gear is not engaged and the aircraft continues in flight.

C

Career Crewmember (also known as Career Enlisted Flyer). A member of the Navy enlisted aviation community rating (AD, AE, AM, AMH, AME, AMS, AO, AT, AV, AW, PR, IT (TACAMO only), or AZ (TAR only)) holding a 78XX, 82XX, or 94XX NEC; or is in a formal training pipeline leading to the award of those NECs, and is detailed by PERS-404E or NRPC-417. Career Enlisted Flyers are crewmembers who are primarily detailed throughout their career into flying billets. Career Enlisted Flyers receive either continuous or conditional Career Enlisted Flyer Incentive Pay (CEFIP) and not Hazardous Duty Incentive Pay (HDIP) for aerial flight.

Chemical, Biological Radiological, or Nuclear Defense (CBRND). Defensive measures taken against the effects of a chemical, biological, or a nuclear weapon attack.

Civil Aircraft Operations. Aircraft operations other than those deemed as Public Aircraft Operations.

Civilian Non-DOD Government Employee. Individual could be with other Federal Government agency, state, county, or local government, etc., or an individual not with any government agency but whose activities benefit the general public at large. Firefighters and in-flight medical services are examples.

Combatant Commander. A commander of one of the unified or specified combatant commands established by the President.

Contract Simulator Instructor (CSI). Contractor or Civil Service personnel designated by a service training agency or CNAF/CMC as a simulator instructor. A Contractor Simulator Instructor may be designated as an assistant NATOPS instructor and/or a NATOPS Instrument Evaluator by the respective Type Wing/Marine Air Group commander. CSIs who maintain these designations shall receive a NATOPS and instrument standardization evaluation annually by an appropriate NATOPS evaluator.

Control (Radar).

- a. Advisory. The tactical control of aircraft by a designated control unit in which the pilot receives directions and recommendations. Aircraft commanders are not relieved of responsibility for their own safety and navigation.
- b. Close. The tactical control of aircraft by a designated control unit, whereby the pilot receives orders affecting aircraft movements. The pilot will not deviate from controller instructions unless given permission or unless unusual circumstances require immediate action for the safety of the flight. In either case, the pilot will inform the controller of the action taken. This type of control requires two-way radio communication and radar contact. The controller is responsible for the safe separation of the aircraft, and the pilot must be informed whenever the aircraft is not held on the radarscope for periods in excess of

1 minute or five sweeps of the radar and, as a result, is being dead reckoned. The ultimate safety of the aircraft is the responsibility of the pilot.

- c. Positive. The tactical control of aircraft by a designated control unit, whereby the pilot receives orders affecting aircraft movements that transfer responsibility for the safe navigation of the aircraft to the unit issuing such orders. The ultimate safety of the aircraft is the responsibility of the pilot.

Controlling Custodian. The command exercising administrative control of assignment, employment, and logistic support of aircraft. Controlling custodians are identified in COMNAVAIRFORINST 4790.2A.

Conversion Mode. Flight operations with the nacelles set between 74° and 5° are considered to be in CONV mode. (Constant nacelle settings between 5° and 1° are not selectable by the pilot.)

Crew Resource Management (CRM). The use of specifically defined behavioral skills as an integral part of every flight to improve mission effectiveness by minimizing crew preventable errors, maximizing crew coordination, and optimizing risk management.

Cross-Country Flight. A flight that either does not remain in the local flying area or remains in the local flying area and terminates at a facility other than an active military facility.

D

Designations. A designation is a one-time occurrence and remains in effect until removed for cause. Commanders shall issue a designation letter to the individual upon the occasion of his/her original designation with appropriate copies for inclusion in his/her NATOPS qualification jacket.

DIFCREW. Duty for enlisted personnel in a flying status involving operational or training flights.

DIFDEN. Duty involving flying denied; duty in a flying status for an officer not involving flying.

DIFOPS. Duty in a flying status for an officer involving operational or training flights.

DIFTEM (USN). Duty in a temporary flying status performing special mission duties as a non-crew member. Enlisted personnel are so ordered in accordance with BUPERINST 1326.4 (series).

Direct Station-to-Station Communications. A means of passing flight progress information between airfields. Communications should be established by one of the following methods:

- a. Voice landline.
- b. Aeronautical Information System (AIS).

E

Enlisted Crewmember (USMC). Enlisted personnel on competent orders to perform duty involving frequent and regular participation in aerial flight as a crewmember.

Enlisted Noncrewmember on Flight Orders (USMC). Enlisted personnel on competent orders to perform duty involving frequent and regular participation in aerial flight who are not performing duties related to the actual operation of the aircraft or associated equipment in the aircraft (i.e., maintenance personnel who perform inflight functions such as installation or troubleshooting of airborne technical equipment (maintenance skins) and VIP support, photo specialists, etc.).

F

Flight.

- a. For operational purposes, a flight is one or more aircraft proceeding on a common mission.
- b. For recording and reporting purposes, a flight begins when the aircraft first moves forward on its takeoff run or takes off vertically from rest at any point of support and ends after airborne flight when the aircraft is on the surface and either:
 - (1) The engines are stopped or the aircraft has been on the surface for 5 minutes, whichever comes first.
 - (2) A change is made in the pilot in command.

- c. For helicopters, a flight begins when the aircraft lifts from a rest point or commences ground taxi and ends after airborne flight when the rotors are disengaged or the aircraft has been stationary for 5 minutes with rotors engaged.

Note

Flight time on repetitive evolutions such as field carrier landing practice (FCLP), passenger/cargo stops, and carrier qualifications shall be logged from the time the aircraft takes off until the aircraft has been on the surface for 5 minutes after each evolution flown (i.e., three sorties of 55 minutes actual air time interspersed with two 20-minute ground periods for refueling or passenger/ cargo transfer will be logged as 3.0 hours of flight time).

Flight Clearance. A flight clearance provides temporary flight operating limits for an aviation system operating in a nonstandard configuration or to a nonstandard envelope, pending issuance of the technical directive or change to the NATOPS, NATIP, or tactical manuals. A flight clearance is a temporary airworthiness approval from COMNAVAIRSYSCOM.

Flight Crew. Personnel whose presence is required on board a manned aircraft or at a control station for UAS to perform crew functions in support of the assigned mission (e.g., pilot, copilot, navigator, flight engineer, crew chief, air observer, special crew, trainee, etc.).

Flight Support Personnel. Personnel immediately involved in the maintenance, fueling, towing/moving, start-up, taxi, or launch and recovery of aircraft including, but not limited to, taxi directors, catapult and arresting gear crew, final checkers, landing signal enlisted (LSEs), aircraft maintenance personnel and aircraft move crews and directors.

Flight Time. The elapsed time computed in accordance with the definition of flight. Flight time is logged in hours and tenths of hours and is creditable to the aircraft, personnel aboard, and equipment.

Formation Flight. A flight of more than one aircraft operating by prior arrangement as a single aircraft with regard to altitude, navigation, and position reporting, and where separation between aircraft within the flight rests with the pilots in that flight.

H

Hazard. A condition with the potential to cause personal injury or death, property damage, or mission degradation.

I

Individual Flight Time. The total pilot time and special crew time creditable to an individual.

Instructor. A naval aviator, naval flight officer, or naval aircrewman designated in writing by competent authority as a flight instructor, NATOPS evaluator, or NATOPS instructor in the aircraft model being flown.

Instructor Time. Individual flight time during which an instructor is required to instruct or evaluate other aeronautically designated personnel or students undergoing a formal flight syllabus.

Instrument Meteorological Conditions. Meteorological conditions expressed in terms of visibility, distance from clouds, and ceiling less than the minimums specified for visual meteorological conditions. IMC conditions exist anytime a visible horizon is not distinguishable.

Instrument Time. The portion of pilot time in either day or night under actual or simulated instrument conditions.

- a. Actual instrument time will be logged by both pilots in a dual/multipiloted aircraft during flight in actual instrument conditions.
- b. Simulated instrument time shall be logged only by the pilot actually manipulating the controls.

Note

NFOs and student NFOs may report actual instrument time if they fly in an aircraft in which they can monitor the pilot instruments and recommend information to the pilot during actual instrument conditions.

J

Joint Service Battlestaff Personnel Embarked on Naval Aircraft. Personnel of all services serving as Battlestaff crewmembers on board Navy E-6 aircraft conducting airborne strategic communications.

L

Landing. A return to the surface; landings include touch and go (providing the landing gear touches the surface), bolter, forced, or crash.

Note

Terms of control terminology such as immediately, possible, and practicable refer to the degree of urgency intended in the message:

- a. Land immediately — Execute a landing without delay.
- b. Land as soon as possible — Land at the first site at which a safe landing can be made.
- c. Land as soon as practicable — Extended flight is not recommended. The landing site and duration of flight is at the discretion of the pilot in command.

Local Flight. A flight that remains within the local flying area and terminates at either the same facility or another military facility with which the originating station has direct station-to-station communications.

Local Flying Area. That area in the vicinity of an air installation in which locally-based aircraft can operate during an average/typical sorties flight time. The local flying area shall not exceed 350 miles from an air installation and be designated as such in the Air Operations Manual by the Commanding Officer. In so far as practicable, local flying areas shall be bounded by prominent terrain features and/or air navigation aid radials/distances.

M

Mile. All distances referred to in this instruction are nautical miles unless otherwise specified.

Mission Commander Time. Flight time during which an individual, designated as a qualified mission commander in the aircraft model being flown, is serving as the mission commander. Mission commander time is a measure of command experience rather than flight experience.

Multipiloted Aircraft. Any aircraft having two sets of flight controls and instruments and operated by two pilots, both of who meet the requirements of the NATOPS manual for that model aircraft.

N

Naval Aircraft. For the purposes of this instruction, those aircraft accepted into the naval aircraft inventory reporting system, pre-accepted aircraft, and public use aircraft operated exclusively by or for the Navy.

Naval Aircrewman. A designation for enlisted personnel who have met the requirements for qualification and have been so certified in accordance with [paragraph 12.9](#) of this instruction.

Naval Aviation Shore Facility. A facility at which an active airfield exists and is either owned, operated, or controlled by the Navy or Marine Corps.

Naval Aviator. A Naval Aviator (NA) is an officer or warrant officer in the United States Navy or Marine Corps that is qualified as a pilot.

Naval Flight Officer. A Naval Flight Officer (NFO) is an officer in the United States Navy or Marine Corps that specializes in airborne weapons and sensor systems. They may perform many co-pilot functions and operate the advanced systems on board in multi-crew naval aircraft, depending on the type of aircraft, and may also act as the overall mission commander and tactical coordinator of multiple air assets during a mission.

Night Time. The portion of pilot time during darkness (i.e., between the official time of sunset and sunrise (on the surface below the aircraft in flight), regardless of whether visual or instrument conditions exist).

O

Oceanic. A situation where an aircraft has no radio communication and is greater than 250 nm from the nearest navaid.

Officer in Tactical Command. The senior officer present eligible to assume command, or the officer to whom he has delegated tactical command.

Official Business. The necessity to contact personnel, units, or organizations for the purpose of conducting transactions in the service of and in the interest of the United States Government. This definition does not authorize the use of official business only airfields, their services, or other items attendant to itinerant operations when making en route stops while proceeding to an airfield at which official business is to be conducted. Official business only restrictions do not preclude the use of the facility as an alternate during instrument flight rule (IFR) conditions.

Operational Flying. (See [paragraph 11.2](#) for definition and application.)

Operational Necessity. A mission associated with war or peacetime operations in which the consequences of an action justify accepting the risk of loss of aircraft and crew.

Operational Risk Management. The process of dealing with the risk associated with military operations, which include risk assessment, risk decision making and implementation of effective risk controls.

Orientation Flight. A continuous-flight in DOD aircraft performed within the local flying area and terminating at the point of origin intended to further the understanding of particular programs concerning the roles and missions of the Department of Defense.

P

Passenger. An individual who is not part of the aircrew traveling in an aircraft designed or normally configured for passenger (non-aircrew) carrying capability on a point-to-point flight.

Pathfinder. An aircraft whose primary mission is to assist tactical aircraft with communication or navigation of flights over regions where normal tactical aircraft navigation/communication equipment is unusable.

Pilot in Command. The pilot assigned responsibility for safe and orderly conduct of the flight.

Pilot Time. The flight time credited to a designated aviator, student naval aviator, student/designated naval flight surgeon, student/designated aerospace physiologist, or student/designated aerospace experimental psychologist assigned to duty involving flying. Pilot time includes all time credited as first pilot and copilot. Pilot time is intended to be a record of active participation in the control of an aircraft. Pilot time will be credited to the individual actually earning it regardless of rank, billet, age, or level of experience.

- a. **First Pilot Time.** The portion of pilot time during which an individual is positioned with access to the flight controls and is exercising principal active control of the aircraft.
- b. **Copilot Time.** The portion of pilot time while assisting the pilot exercising principal active control of a multipiloted aircraft during which the copilot is positioned with access to and is immediately ready to operate the flight controls; or, in those aircraft with only one set of flight controls, that portion of flight time while instructing the pilot who is exercising principal active control when the designated instructor is positioned so that pilot and aircraft instruments can be observed. Aeronautically designated personnel may log CPT while performing copilot duties as required by the aircraft mission.

Pilot Under Instruction. A designated aviator under instruction.

Pre-Accepted Aircraft. Those aircraft under development or in production for the Navy which have not yet been accepted into the naval aircraft inventory via DD 250.

Project Specialist. A non-aeronautically designated individual (military or civilian) embarked in a government aircraft for the purpose of operating aircraft systems, operating specially designed equipment, or observing aircraft or crew performance, etc. when required in conjunction with assigned duties or contractual responsibilities. Project specialists are not responsible for normal aircrew duties. This category is not appropriate for those completing orientation flights or for midshipmen.

Public Aircraft Operations. In general, the U.S. Armed Forces consider an aircraft operation “Public” when the aircraft is owned by the Armed Forces, or is used by the Armed Forces and operates outside of the purview of its FAA airworthiness certificate (e.g. configuration, operational use, flight rules, or maintenance). See 49 U.S.C. §§ 40102 (A)

(41) and 40125. Any command considering employing contracted aircraft services that may be conducting PAO is encouraged to contact COMNAVAIRSYSCOM (AIR 4.0P Airworthiness Directorate) for assistance.

Q

Qualified in Model. A designation that indicates the minimum requirements for qualification in a specific crew position, as set forth in the appropriate NATOPS manual, have been attained. Such designations are a one-time occurrence (per unit/command tour) and remain in effect until removed for cause. Annual NATOPS evaluations should not be confused with or combined with these designations. If specific aircraft model NATOPS guidance is lacking, an individual shall be considered qualified in model for specific crew position when so designated by the reporting custodian.

R

Reporting Custodian. An organizational unit of the lowest echelon of command accepting responsibility (involving accountability to CNO) for aircraft as designated either by CNO or by the controlling custodian of the aircraft.

Risk. An expression of possible loss in terms of severity and probability.

Risk Assessment. The process of detecting hazards and assessing associated risks.

S

Simulated Instrument Approach. An instrument approach flown under simulated instrument conditions.

Simulated Instrument Conditions. Conditions external to the aircraft in flight are visual meteorological conditions (VMC), but pilot vision is limited primarily to the interior of the aircraft.

Single-Piloted Aircraft. Any aircraft that has only one set of flight controls or a tandem cockpit, or any aircraft that has two sets of flight controls and instruments and is being operated by only one pilot who meets the requirements of the NATOPS manual for that model aircraft.

Special Crew Time. The portion of flight time accrued while not acting as first pilot or copilot, but otherwise serving as a member of the authorized crew complement of an aircraft or as a student in flight training.

Special Operations Forces Personnel. Herein referred to as SPECOPS. Personnel who are required to conduct special operations such as high altitude parachuting from military aircraft (SEALS, ANGLICO, RECON, etc.).

Standardization Evaluation. An evaluation conducted by the NATOPS evaluator for the purpose of measuring the knowledge and instructing capabilities of a NATOPS Instructor or Evaluator. This evaluation may be performed coincident with any annual NATOPS evaluation.

Stereo Route. Routinely used route of flight established by users and ARTCC identified by a coded name. These routes simplify flight plan handling and communications.

Student Naval Aviator (Student Pilot). An individual undergoing training who is not designated as a naval aviator.

T

Tilt-rotor. Aircraft type capable of rotor-borne and wing-borne flight (e.g., MV-22).

Trip. A consecutive series of flights by the same aircraft with the same general purpose of flight (with regard to the aircraft only), pilot in command, and transaction code (i.e., ship operations or shore operations) from point of original departure to destination.

U

Unmanned Aircraft (UA). A rotary, fixed-wing, or lighter-than-air aircraft which is capable of flight without an on-board crew. UA can be operated autonomously or remotely, can be expendable or recoverable, and can carry a lethal or nonlethal payload. Ballistic or semi-ballistic vehicles, cruise missiles, artillery projectiles, torpedoes, mines, satellites, and unattended sensors (with no form of propulsion) are not considered unmanned vehicles. Unmanned aircraft are the primary component of unmanned aircraft systems (UAS).

Unmanned Aircraft System(s) (UAS). The system, whose components include the necessary equipment, data communication links, and personnel to control and employ an unmanned aircraft (UA).

Unmanned Aerial Vehicle. A legacy term replaced by Unmanned Aircraft (UA) or Unmanned Aircraft System (UAS) as applicable.

V

Very Important Persons. VIPs are defined as flag officers, DOD officials equal to or senior to flag officers, high-profile public figures, elected members of Congress, etc.

Visual Meteorological Conditions.

Meteorological conditions expressed in terms of visibility, cloud distance, and ceiling that are equal to or better than specified minimums. Basic weather conditions prescribed for flight under visual flight rules (VFR). (Refer to Chapter 5.)

VOD. For the purposes of this instruction, all helicopter and tilt-rotor aircraft that have the capability to deliver passengers or cargo.

LIST OF ABBREVIATIONS/ACRONYMS

A

ABI.	Aviation billet indicator.	AMCM.	Airborne mine countermeasures.
ACFT CMDR.	Aircraft commander.	AMDD.	Aeromedical Dual Designator.
ACIP.	Aviation career incentive pay.	AMO.	Aviation medical officer.
ACM.	Air combat maneuvers.	AMSO.	Aeromedical Safety Officer.
ACP.	Allied communication publication.	AOA.	Angle of attack.
ACT.	Aircraft commander time.	AP.	Area planning.
ACTS.	Assignment, classification and travel system.	APMSE.	NAVAIR Assistant Program Manager for Systems Engineering.
ADIZ.	Air defense identification zone.	APT.	Aircrew procedure trainer; Aviation physiology technician.
ADRL.	Automatic distribution requirements listing.	AQD.	Additional qualification designation.
ADW.	Aviation data warehouse.	ARTCC.	Air route traffic control center.
AEW.	Airborne early warning.	ASAP.	Aviation safety awareness program.
AFCS.	Automatic flight control system.	ASED.	Aviation service entry date.
AFRL.	Air Force Research Laboratory.	ASEP.	Aircrew survivability enhancement program.
A/G.	Air-to-ground.	ASI.	Aviation status indicator.
AGL.	Above ground level.	ASR.	Surveillance radar approach.
AI.	Air intelligence; Air intercept.	ASTC.	Aviation Survival Training Center.
AIA.	Aircraft inspection and acceptance.	ASW.	Antisubmarine warfare.
AIRS.	Airworthiness issue resolution system; Aircraft inventory reporting system.	ATC.	Air traffic control.
AIS.	Aeronautical Information System.	ATCAA.	Air traffic control assigned airspace.
ALS.	Approach lighting system.	ATP.	Allied tactical publication.
ALSS.	Aviation life support system.	AV-3M.	Aviation maintenance and material management.
ALTRV.	Altitude reservation.	AVO.	Air vehicle operator.
AMB.	Aviation mishap board.	AVOC.	Airfield Vehicle Operators Course.
		AVOPS.	Aviation Operations Officer.

B

BAM. Basic aircraft maneuvers.

BFM. Basic fighter maneuvers.

BUMED. Bureau of Medicine and Surgery.

BuNo. Bureau number.

BUQ. Basic UAS qualification.

C

CASREP. Casualty report.

CB. Chemical, biological.

CBR. Chemical, biological, and radiological.

CBRND. Chemical, biological, radiological, or nuclear defense.

CCA. Carrier-controlled approach.

CG FOURTH MAW. Commanding General, 4th Marine Air Wing.

CEFIP. Career Enlisted Flyer Incentive Pay.

CFET. Centrifuge-based Flight Environment Training.

CFR. Code of Federal Regulations.

COMUSNAVEUR. Commander, U.S. Naval Forces Europe.

COMUSNAVCENT. Commander, U.S. Naval Forces Central.

COMUSNAVSO. Commander, U.S. Naval Forces South.

CMC. Commandant of the Marine Corps.

CNAF. Commander, Naval Air Forces.

CNAL. Commander, Naval Air Force, U.S. Atlantic Fleet.

CNAP. Commander, Naval Air Force, U.S. Pacific Fleet.

CNATRA. Chief of Naval Air Training.

CNI. Communication, navigation, identification.

CNO. Chief of Naval Operations.

CO. Commanding Officer.

COD. Carrier on-board delivery.

COG. Cognizant command.

COMFLTFORCOM. Commander, Fleet Forces Command.

COMMARFORCOM. Commander, U.S. Marine Forces Command.

COMMARFORPAC. Commander, U.S. Marine Forces, Pacific.

COMNAVAIRFOR. Commander, Naval Air Forces.

COMNAVAIRFORES. Commander, Naval Air Force Reserve.

COMNAVAIRLANT. Commander, Naval Air Force, U.S. Atlantic Fleet.

COMNAVAIRPAC. Commander, Naval Air Force, U.S. Pacific Fleet.

COMNAVAIRSYSCOM. Commander, Naval Air Systems Command.

COMNAVSAFECEN. Commander, Naval Safety Center.

COMSEVENTHFLT. Commander, Seventh Fleet.

COMSIXTHFLT. Commander, Sixth Fleet.

COMUSNAVCENT. Commander, U.S. Naval Forces Central Command.

COMUSNAVEUR. Commander, U.S. Naval Forces Europe.

COMUSNAVSO. Commander, U.S. Naval Forces Southern Command.

CONUS. Continental United States.

CPT. Copilot time.

CR. Canned Route.

CRM. Crew Resource Management.

CSI. Contractor Simulator Instructor.

CTF. Commander Task Force.

CVW. Carrier Air Wing.

D

DAFIF. Digital Aircraft Flight Information File.

DCF. Document control form.

DCMA. Defense Contract Management Agency.

DEWIZ. Defense early warning identification zone.

DH. Decision height.

DIFCREW. Duty involving flying, crewman.

DIFDEN. Duty in a flying status not involving flying.

DIFOPS. Duty in a flying status involving operational or training flights.

DIFTECH. Duty involved flying as a technical observer.

DIFTEM. Personnel under training to become crewmembers.

DME. Distances measuring equipment.

DNEC. Distributive naval enlisted classification.

DOD. Department of Defense.

DON. Department of the Navy.

DONI. Department of the Navy issuances.

DP. Departure procedure.

DSN. Defense switched network.

DUAT. Direct user access terminal.

E

ECM. Electronic countermeasures.

ESCAT. Emergency Security Control of Air Traffic.

ETA. Estimated time of arrival.

ETD. Estimated time of departure.

ETE. Estimated time en route.

F

F/W. Fixed wing.

FAA. Federal Aviation Administration.

FAAO. FAA Order.

FACSFAC. Fleet area control and surveillance facility.

FAILSAFE. Fleet air introduction/liaison of survival aircrew flight equipment.

FAR. Federal Aviation Regulation.

FCF. Functional checkflight.

FCLP. Field carrier landing practice.

FDLP. Flight deck landing practice.

FFPB. Field Flight Performance Board.

FL. Flight level.

FLIP. Flight information publication.

FLIR. Forward looking infrared.

FMC. Full mission capable.

FMF. Fleet Marine Force.

CNAF M-3710.7

FMS. Foreign military sales.

FNAEB. Field Naval Aviator Evaluation Board.

FOD. Foreign object damage.

FPC. Flight purpose code.

FPT. First pilot time.

FRC. Fleet Readiness Center.

FREDS. Flight readiness evaluation data system.

FRS. Fleet Replacement Squadron.

FS. Flight surgeon.

FSS. Flight service station.

FSSB. Flight Status Selection Board.

FTI. Flight Training Instruction.

FWB. Flight Weather Briefer.

FXP. Fleet exercise publication.

FYTD. Fiscal year to date.

G

GCA. Ground-controlled approach.

GCI. Ground-controlled intercept.

GCS. Ground control station.

GLOC. G-loss of consciousness.

GMVO. Ground maintenance vehicle operator.

GPC. General purpose code.

GPS. Global positioning system.

GSA. General Services Administration.

ORIGINAL

H

HABD. Helicopter aircrew breathing device.

HAP. High-altitude parachute.

HAT. Height above touchdown.

HDIP. Hazardous duty incentive pay.

I

IAF. Initial approach fix.

IAW. In accordance with.

IC. Interim Change.

ICAO. International Civil Aviation Organization.

ICE-T. Memorization acronym for indicated, calibrated, equivalent, and true airspeeds.

IDC. Independent duty corpsman.

IAF. Initial approach fix.

IFARS. Individual flight activity reporting system.

IFC. Interim flight clearance.

IFF. Identification friend or foe.

IFR. Instrument flight rules.

ILS. Instrument landing system.

IMC. Instrument meteorological conditions.

IMR. Individual master roster.

IT. Instructor time.

IUT. Instructor under training.

J

JAGMAN. Manual of the Judge Advocate General.

JANAP. Joint Army, Navy, Air Force publication.

JATO. Jet assisted takeoff.

JHMCS. Joint Helmet Mounted Cueing System.

JUMPS. Joint uniform military pay system.

K

KIAS. Knots indicated airspeed.

L

LATT. Low altitude terrain tactics

LBFS. Local board of flight surgeons.

LEO. Law enforcement official.

LEP. Laser eye protection.

LIMDU. Limited duty.

LOA. Letter of agreement.

LOL. Loss of link.

LOP. Letter of promulgation.

LOS. Line of sight.

LPU. Life preserver unit.

LSO. Landing signal officer.

M

MAG. Marine aircraft group.

MANMED. Manual of the Medical Department.

MAP. Military assistance program; missed approach point.

MARFORCOM. U.S. Marine Forces Command.

MARSA. Military assumes responsibility for separation of aircraft.

MAW. Marine Air Wing.

MCAS. Marine Corps Air Station.

MCO. Marine Corps Order.

MCIEAST. Marine Corps Installations East.

MCIWEST. Marine Corps Installations West.

MCS. Mission Control Station (UAS).

MDA. Minimum descent altitude.

MDS. Maintenance data system.

MEDEVAC. Medical evacuation.

METS. Modular Egress Training System.

MFOQA. Military flight operations quality assurance.

MIFAR. Monthly individual flight activity report.

MIM. Maintenance instruction manual.

MITO. Minimum interval takeoff.

MMU. Model Manager Unit.

MOA. Military operating areas.

MOF. Month(s) operational flying.

MOS. Military occupational specialty.

MPO. Mission payload operator.

MPT&E Co E. Navy Medicine Manpower Personnel Training and Education Command Council on Occupational Education.

M-SHARP. Marine Corps Sierra Hotel Aircraft Readiness Program.

MSDD. Multi-station disorientation demonstrator.

MSL. Mean sea level.

MSN CDR. Mission Commander.

MTR. Military training route.

N

NA. Naval aviator; Naval air systems command; not applicable.

NAC. Naval aircrewman.

CNAF M-3710.7

NACCS. Naval Air Crew Candidate School.

NAFC. Naval Aviation Forecast Center.

NALCOMIS. Naval Aviation Logistics Command Management Information Systems.

NALDA. Naval Aviation Logistics Data Activity.

NALIS. Navy Logistics Information System.

NAMI. Naval Aerospace Medical Institute.

NAMT. Naval air maintenance trainer.

NAP. Naval aerospace physiologist.

NAS. Naval air station; National airspace.

NASA. National Aeronautics and Space Administration.

NASTP. Naval Aviation Survival Training Program.

NATEC. Naval Air Technical Data and Engineering Service Command.

NATIP. Naval Aviation Technical Information Product.

NATO. North Atlantic Treaty Organization.

NATOPS. Naval Air Training and Operating Procedures Standardization.

NAVAEROMEDINST. Naval Aeromedical Institute (NAMI).

NAVAID. Navigation aid.

NAVAVSCOLSCOM. Naval Aviation Schools Command.

NAVFIG. Naval Flight Information Group.

NAVFLIRS. Naval Flight Record Subsystem.

NAVMETOCCOM. Naval Meteorology and Oceanography Command.

NAVOCEANO. Naval Oceanographic Office.

NAVPERSCOM. Navy Personnel Command.

NAVREP. Navy representative.

NAVRESPERSCEN. Naval Reserve Personnel Center.

NCOIC. Noncommissioned officer in charge.

NDB. Non-directional beacon.

NEC. Naval enlisted classification.

NETC. Naval Education and Training Command.

NFM. NATOPS flight manual.

NFO. Naval flight officer.

NGA. National Geospatial Agency.

NI. NATOPS instructor.

NITE. Night imaging and threat evaluation.

NJROTC. Naval Reserve Junior Officer Training Corps.

nm. Nautical mile.

NMC. Not mission capable.

NMCS. Not mission capable-supply.

NMCM. Not mission capable-maintenance.

NOBC. Naval officer billet code.

NOE. Nap of the Earth.

NOMI. Naval Operational Medicine Institute.

NORDO. No-radio.

NOS. National Oceanographic Service.

NOTAL. Not sent to all.

NOTAM. Notice to airmen.

NPA. Non-precision approach.

NPQ. Not physically qualified.

NSTI. Naval Survival Training Institute.

NTTP. Naval Tactics, Techniques, and Procedures publication.

NVCD. Night vision cueing device.

NVD. Night vision device.

NWP. Naval warfare publication.

O

OAT. Outside air temperature.

OCF. Out of control flight.

OCONUS. Outside of the continental US.

ODCR. Officer data control report.

OEM. Original equipment manufacturer.

OFT. Operational flight trainer.

OIC. Officer in charge.

OLF. Outlying field.

OMA. Operational Maintenance Activity.

OMB. Office of Management and Budget.

ORG. Organization.

ORI. Operational readiness inspection.

ORM. Operational risk management.

OSD. Office of the Secretary of Defense.

OTC. Officer in tactical command; Over the counter.

OT&E. Operational test and evaluation.

P

PALS. Precision approach and landing system.

PAR. Precision approach radar.

PCS. Permanent change of station.

PEP. Personnel exchange program.

PIC. Pilot in command.

PMC. Partial mission capable.

POC. Point of contact.

POR. Program of Record.

PPR. Prior permission required.

PPS. Precise Positioning Service.

PQM. Pilot qualified in model.

PQS. Personnel qualification standard.

PR. Parachute rigger.

Q

QRG. Quick Reference Guide.

R

RAC. Replacement aircrew.

RDO. Runway duty officer.

RDT&E. Research, development, test, and evaluation.

RNAV. Radar navigation.

RNP. Required Navigation Performance.

ROBD. Reduced oxygen breathing device.

RON. Remain overnight.

ROTC. Reserve Officer Training Corps.

RSS. Rescue Swimmer School.

RSSMM. Rescue Swimmer School model manager.

RSSTP. Rescue Swimmer School training program.

RTO. Range training officer.

CNAF M-3710.7

RUC. Reporting unit code.

RVR. Runway visual range.

RVSM. Reduced vertical separation minima.

S

SA. Situational awareness.

SAD. Senior air director.

SAR. Search and rescue.

SARMM. Search and rescue model manager.

SARP. Standards and recommended practices.

SCA. Self-contained approach.

SCATANA.K Security control of air traffic and air navigation aide.

SCT. Special crew time.

SEBD. Supplemental emergency breathing device.

SELRES. Selective Reserve.

SERE. Survival, evasion, resistance to interrogation and escape.

SFA. Single frequency approach.

SHARP. Sierra Hotel Aviation Readiness Program.

SIF. Selective identification feature.

SIGMET. Significant meteorological information advisory.

SIM. Simulated.

SMCR. Selected Marine Corps Reserve.

SOP. Standard operating procedure.

SPC. Specific purpose code; Storm prediction center.

SPECOPS. Special Operations Forces personnel and/or missions.

SPS. Standard positioning device.

STANAG. Standardization agreement.

STAR. Standard terminal arrival notice.

STOL. Short takeoff and landing.

SUA. Special use airspace.

T

T&R. Training and readiness.

TACAN. Tactical air navigation.

TACTS. Tactical aircrew combat training system.

TAR. Training air reserve.

TAD. Temporary additional duty.

TBA. To be assigned.

TCAS. Traffic Alert and Collision Avoidance System.

TDIP. Technical data indoctrination package.

TERPS. Terminal instrument procedures.

TMR. Total mission requirements.

T/M/S. Type/model/series.

TO. Table of organization.

TR. Training rules.

TRACOM. Training command.

TTT. Team Tactics Trainer.

TYCOM. Type Commander. The aviation TYCOMs are CNAF and CNAL.

U

UA. Unmanned Aircraft.

UAC. UAS aircraft commander.

UAS. Unmanned Aircraft System(s).

UASC. UAS crewmember.

UHF. Ultra high frequency.

UIC. Unit identification code.

UMC. UAS mission commander.

USG. US Government.

USW. Undersea warfare.

UT. Under training.

UTC. Coordinated Universal Time.

V

VFR. Visual flight rules.

VHF. Very high frequency.

VIP. Very important person.

VMC. Visual meteorological conditions.

VOD. Vertical on-board delivery.

VOR. VHF Omni-Directional Range.

VR. Transport (aircraft/mission); VFR Military Training Route.

V/STOL. Vertical/short takeoff and landing.

W

WST. Weapon system trainer.

WW. Severe weather watch.

CHAPTER 1

Introduction

1.1 GENERAL

OPNAVINST 3710.7 establishes the Naval Air Training and Operating Procedures Standardization (NATOPS) program. This manual contains detailed policies and procedures in support of OPNAVINST 3710.7. The NATOPS program is a positive approach towards improving combat readiness and achieving a substantial reduction in aircraft mishaps. This manual issues policy and procedural guidance of the Chief of Naval Operations (CNO) that is applicable to all NATOPS users.

1.1.1 Purpose and Scope

- a. This manual prescribes general flight and operating instructions and procedures applicable to the operation of all naval aircraft and related activities. This manual is not intended to cover every contingency that may arise and every rule of safety and good practice. To achieve maximum value, the contents of all directives cited must be studied and understood. Routine interpretation and procedural questions should be referred to appropriate type wing/type command or MAG/MAW NATOPS offices for resolution prior to referral to a cognizant command (COG)/Naval Advisory Group (NAG). Where the need arises, special instructions or waivers will be issued by COMNAVAIRFOR or CMC as applicable.
- b. Directives cited in the text of this manual are identified by the basic number, for example OPNAVINST 8000.16B will be identified as OPNAVINST 8000.16. [Appendix C](#) lists the specific version of each directive used in the preparation of the current manual. It is the responsibility of the user to determine the current status of any directive being used.
- c. In the tactical environment, military exigency may require on-site deviations from instructions/procedures contained here. The existing risk of deviation must continually be weighed against the benefit of deviating from this manual. Deviation from specified flight and operating instructions is authorized in emergency situations when, in the judgment of the pilot in command, safety justifies such a deviation.
- d. It is often not feasible to completely specify all situations or circumstances under which provisions of this manual shall apply; therefore, wording such as “normally,” “etc.,” “usually,” and “such as” is employed. Words or clauses of that type shall not be used as loopholes nor shall they be expanded to include a maneuver, situation, or circumstance that should not be performed or encountered by the aircraft in question.
- e. To increase combat readiness and improve flight safety, the scope and operation of the NATOPS program, conduct of NATOPS evaluations, NATOPS change recommendation procedures, and NATOPS Review procedures are discussed in [Chapter 2](#).

1.1.2 Submission of NATOPS Change Recommendations

Recommended changes to this and other NATOPS publications may be submitted by anyone in accordance with [Chapter 2](#) of this manual. The preferred and most efficient means for submission of change recommendations for both this manual and the aircraft and general series NATOPS publications is to do so via the Airworthiness Issue Resolution System (AIRS) program found on the NATOPS page of the Airworthiness website, <https://airworthiness.navair.navy.mil>. When access to AIRS is not available, recommended urgent and priority changes should be submitted by Naval Message. Submit recommended routine changes to this manual to Commander Naval Air Forces (N455), NAS North Island, P.O. Box 357051, San Diego, CA 92135-7051 via Form OPNAV 3710/6 only when unable to do so via AIRS.

1.1.3 Change Symbols

Revised text is indicated by a black vertical line in either margin of the page, adjacent to the affected text, like the one printed next to this paragraph. The change symbol identifies the addition of new information, a changed procedure, the correction of an error, or a rephrasing of the previous material.

1.1.4 Areas of Responsibility

Figure 1-1 delineates administrative areas of responsibility within this manual. Questions about policy content should be referred to the appropriate command listed in **Figure 1-1**. For waivers to the requirements set forth in this manual, refer to **paragraph 2.2.3 b**.

ORGANIZATION	CHAPTER
COMNAVAIRFOR (N455)	Chapters 1, 3–7, 9–14, and Appendices A–D and F–N
COMNAVAIRSYSCOM (AIR-4.0P)	2
BUMED (M3B3)	8, Appendix E

Figure 1-1. CNAF M-3710.7 Areas of Responsibility

1.1.5 How To Obtain Copies

1.1.5.1 CNAF M-3710.7

Distribution of this manual is by electronic means only. Electronic copies of the revisions, changes and interim changes to this manual can be found in the following locations:

- Unclassified SECNAV and OPNAV directives are at the Department of the Navy Issuances (DONI) website: <http://doni.documenservices.dla.mil/default.aspx>.
- Airworthiness Website — <https://airworthiness.navair.navy.mil>.
- NATEC website — <https://mynatec.navair.navy.mil>.

1.1.5.2 NATOPS Publications

- Automatic Distribution.** Automatic distribution of individual NATOPS publications are as requested by the individual units in their Automatic Distribution Requirements Listing (ADRL) accounts. Units flying the aircraft will receive paper copies based on requirements determined by the NATOPS Model Manager. Other units will receive distribution via appropriate digital media, whenever available, as determined from the unit's ADRL request.
- Additional Copies.** Those who require paper copies can obtain them from the NATOPS Model Manager unit, whose address is published in the Preface of each NATOPS publication. The name, rank, telephone number, and e-mail address of the NATOPS Program Manager for each publication is contained in the NATOPS Status Report which is a product posted on the Airworthiness website, <https://airworthiness.navair.navy.mil>. Electronic copies of most NATOPS publications are posted in PDF-format on the NATOPS Search page of the NATEC website, <https://mynatec.navair.navy.mil> and Airworthiness website <https://airworthiness.navair.navy.mil>. Active interim change messages are normally posted on the site within five days of their release.

1.2 APPLICATION OF OPERATIONAL RISK MANAGEMENT ANALYSIS

As directed by DODINST 6055.1 and OPNAVINST 3500.39, Operational Risk Management (ORM) analysis principles shall be applied during the planning and execution phases of all military operations and training situations.

1.3 OTHER GOVERNING SOURCES OF INFORMATION

Instructions and procedures contained in this manual are not intended to replace or duplicate the following governing sources.

1.3.1 NATOPS Manuals

NATOPS manuals include the NATOPS flight manuals, which are issued for particular aircraft (e.g., MH-60S NATOPS Flight Manual), and the General Series NATOPS manuals, which are issued for aviation-related activities (e.g., CV NATOPS Manual). They contain specific doctrine and the optimum operating procedures for the aircraft model or aviation activity concerned. Where a NATOPS manual is not issued for a particular model aircraft, appropriate commands shall issue doctrine and procedures locally. Individual aircraft NATOPS requirements should be at least as stringent as those set forth in this manual. If as a result of a NATOPS Review, it is desired to establish a less stringent requirement, approval shall be obtained from COMNAVAIRFOR. Such approval may be requested by submitting a copy of the Review report to COMNAVAIRFOR (N455) and COMNAVAIRSYSCOM (AIR-4.0P) with the item listed as a change requiring further approval in accordance with [Chapter 2](#). When more stringent requirements are issued in this manual, this manual shall govern until specific authority to deviate has been granted by COMNAVAIRFOR (N455).

1.3.2 Local Flying Rules and Instructions

Local flying rules and instructions will be found in regulations issued by the various fleets, forces, naval air stations, and other activities where naval aircraft are based or operated. Navy and Marine Corps Air Stations and other naval aviation shore facilities that routinely conduct flight operations shall supplement this manual with air operations manuals. Guidelines for the preparation of air operations manuals are contained in NAVAIR 00-80T-114 (ATC NATOPS manual).

1.3.3 Federal Aviation Regulations (FAR)

Naval aircraft shall be operated in accordance with applicable provisions of FAR, Part 91, except:

- a. Where this manual prescribes more stringent requirements.
- b. Where exemptions or authorizations issued to the Department of the Navy/DOD permit deviation from FAR.

1.3.3.1 FAR Exemptions

Users shall determine the expiration date, full scope and restrictions of an exemption prior to exercising it. Exemptions to FARs applicable to DOD aircraft may be viewed on the FAA Automated Exemption System (AES) website, <http://aes.faa.gov>, using petitioner as “Department of Defense” or “Department of the Navy” for USN and USMC exemptions and consulting the AES User Manual as needed. Exemptions/authorizations which are currently on file that allow deviation from FAR Part 91 include:

- a. Section 91.117 (Aircraft Speed). Operation of naval aircraft at speeds in excess of limits imposed by section 91.117 shall be governed by [paragraph 5.1.4](#) of this manual.
- b. Section 91.121 (Altimeter Settings). Allows the use of the local altimeter setting when conducting high-speed tactical maneuvers that include rapid transits of Flight Level 180. (Exemption 2861A, non-expiring).
- c. Section 91.135 (Operations in Class A Airspace). Authorizes USN undergraduate student aviators to conduct solo flight in Class A airspace without an instrument rating.
- d. Section 91.159 (a) (VFR Cruising Altitude or Flight Level). Allows operations at altitudes other than those prescribed by section 91.159 (a) while engaged in drug interdiction operations, only to the extent necessary to

CNAF M-3710.7

obtain positive identification of a suspect aircraft and maintain visual contact with that aircraft, provided the aircraft has a dedicated on-board observer (other than the pilot) to watch for other air traffic, and the aircraft has an operating transponder with Mode C. (Exemption 5100K, expires 9/30/2015.)

- e. Section 91.169 (b) and (c) (Alternate Airport Requirements). Alternate airport requirements and alternate airport weather criteria for clearance of flights to be conducted under IFR shall be specified in [paragraph 4.8.4.2](#) of this manual. (Exemption 30B, non-expiring).
- f. Section 91.179 (b) (1) (IFR Cruising Altitude or Flight Level). Exemption from the altitudes to be maintained in uncontrolled airspace has been granted to the extent necessary to conduct military training route (MTR) training. Policies and procedures for the conduct of MTRs are contained in FAAO JO 7610.4, Special Operations, and in FLIP Area Planning AP/1B. (Exemption 2396, non-expiring).
- g. Section 91.209 (a) (Aircraft Lights). An exemption has been granted to DOD aircraft engaged in drug interdiction flights provided the aircraft has a dedicated on-board observer plus an additional resource capable of detecting all aircraft operating in the vicinity of the DOD aircraft. (Exemption 5100K, expires 9/30/2015.)
- h. Sections 91.209(a) and (b) (Aircraft Lights). An exemption for USMC aircraft from 91.209(a) and (b) for flight without lighted aircraft position lights in order to conduct night vision device flight training in USMC helicopters. (Exemption 8028D, expires 04/30/2015.)

1.3.4 DOD Flight Information Publications (FLIPs) (NOTAL) and Notices to Airmen (NOTAMs) (NOTAL)

The procedures, special notices, and instructions contained in the FLIPs and NOTAMs are mandatory for all pilots flying naval aircraft.

1.3.5 FAA Order JO 7110.65 (Air Traffic Control) (NOTAL)

The FAA order is applicable to air traffic control by Department of Defense (DOD) activities unless individual military service exceptions are noted therein. The applicable procedures shall be used by naval aviation shore facilities when performing air traffic control (ATC) functions. Waivers for deviations from the procedures set forth in FAA order JO 7110.65 may be granted by CNO (N980A). Authority for reduced runway separation for arriving and departing aircraft using the same runway is outlined in [paragraph 6.3.1](#).

1.3.6 NATOPS Air Traffic Control Manual (NAVAIR 00-80T-114)

This manual is applicable to the operation of Navy and Marine Corps air traffic control facilities. Applicable procedures shall be used by shore facilities when performing ATC functions.

1.3.7 DoD Detail Specification for NATOPS Program Technical Publications and Products; Style, Format, and Common Technical Content, MIL-DTL-85025B(AS)

This document contains the style, format, and content standards and requirements for NATOPS publications. It applies to all (paper and electronic media) NATOPS flight manuals, General Series manuals, and checklist publications, and to the other derivative NATOPS-series documents. Persons preparing or modifying NATOPS publications shall be familiar with the specifications and guidelines contained in MIL-DTL-85025B(AS).

1.3.8 Other Instructions

References contained in the manual are listed in [Appendix C](#).

1.4 EXPLANATION OF TERMS

The explanation or definitions of terms and abbreviations commonly used in the aviation community can be found in FAR, Part I, and DOD FLIP General Planning, [Chapter 2](#); and Aeronautical Information Manual (AIM) Pilot/Controller Glossary. No effort to duplicate these terms is intended. Where terms are used in this manual with a different connotation or where definitions are lacking in the above-mentioned publications, the explanations of such terms are included in the Glossary.

1.5 WARNINGS, CAUTIONS, AND NOTES

The following definitions apply to WARNINGS, CAUTIONS, and Notes found throughout this manual.

WARNING

Explanatory information about an operating procedure, practice, or condition, etc., that may result in injury, death, or loss of aircraft if not carefully observed or followed.

CAUTION

Explanatory information about an operating procedure, practice, or condition, etc., that may result in damage to equipment if not carefully observed or followed.

Note

Explanatory information about an operating procedure, practice, or condition, etc., that must be emphasized.

1.6 WORDING

The concept of word usage and intended meaning that has been adhered to in preparing this manual is as follows:

- a. “Shall” is used only when application of a procedure is mandatory.
- b. “Should” is used only when application of a procedure is recommended.
- c. “May” and “need not” are used only when application of a procedure is optional.
- d. “Will” indicates futurity and never indicates any degree of requirement for application of a procedure.
- e. “Land Immediately” means execute a landing without delay.
- f. “Land as Soon as Possible” means land at the first site at which a safe landing can be made.
- g. “Land as Soon as Practicable” means extended flight is not recommended. The landing site and duration of flight are at the discretion of the pilot in command.

